Newman-Penrose Formalism Calculations

Stephani et al., Equation (36.31)

Metric

$$ds^2 = -e^{(2\,\alpha(x,y,t))}\mathrm{d}t \otimes \mathrm{d}t + e^{(2\,b(x,y,t))}\mathrm{d}x \otimes \mathrm{d}x + e^{(2\,b(x,y,t))}\mathrm{d}y \otimes \mathrm{d}y + N\left(z,x,y,t\right)^2 \mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}e^{(\alpha(x,y,t))}\right)\,dt + \left(\frac{1}{2}\sqrt{2}e^{(b(x,y,t))}\right)\,dx$$

$$n = \left(\frac{1}{2}\sqrt{2}e^{(\alpha(x,y,t))}\right)\,dt + \left(-\frac{1}{2}\sqrt{2}e^{(b(x,y,t))}\right)\,dx$$

$$m = \left(\frac{1}{2}\sqrt{2}e^{(b(x,y,t))}\right)\,dy + \left(\frac{1}{2}i\sqrt{2}N\left(z,x,y,t\right)\right)\,dz$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}e^{(b(x,y,t))}\right)\,dy + \left(-\frac{1}{2}i\sqrt{2}N\left(z,x,y,t\right)\right)\,dz$$

Contravariant components

$$l = \left(-\frac{1}{2}\sqrt{2}e^{(-\alpha(x,y,t))}\right)\,dt + \left(\frac{1}{2}\sqrt{2}e^{(-b(x,y,t))}\right)\,dx$$

$$n = \left(-\frac{1}{2}\sqrt{2}e^{(-\alpha(x,y,t))}\right) dt + \left(-\frac{1}{2}\sqrt{2}e^{(-b(x,y,t))}\right) dx$$

$$m = \left(\frac{1}{2}\sqrt{2}e^{(-b(x,y,t))}\right) dy + \left(\frac{i\sqrt{2}}{2\,N\left(z,x,y,t\right)}\right) dz$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}e^{(-b(x,y,t))}\right) dy + \left(-\frac{i\sqrt{2}}{2\,N\left(z,x,y,t\right)}\right) dz$$

Directional Derivatives

$$D = -\frac{1}{2}\sqrt{2}e^{(-\alpha(x,y,t))}\,\partial_t + \frac{1}{2}\sqrt{2}e^{(-b(x,y,t))}\,\partial_x$$

$$\Delta = -\frac{1}{2}\sqrt{2}e^{(-\alpha(x,y,t))}\,\partial_t - \frac{1}{2}\sqrt{2}e^{(-b(x,y,t))}\,\partial_x$$

$$\delta = \frac{1}{2}\sqrt{2}e^{(-b(x,y,t))}\,\partial_y + \frac{i\sqrt{2}}{2\,N\left(z,x,y,t\right)}\,\partial_z$$

$$\overline{\delta} = \frac{1}{2}\sqrt{2}e^{(-b(x,y,t))}\,\partial_y - \frac{i\sqrt{2}}{2\,N\left(z,x,y,t\right)}\,\partial_z$$

Spin Coefficients

$$\rho = -\frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}b\left(x,y,t\right) + \frac{1}{4}\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}b\left(x,y,t\right) - \frac{\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)} + \frac{\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)}$$

$$\sigma = -\frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}b\left(x,y,t\right) + \frac{1}{4}\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}b\left(x,y,t\right) + \frac{\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)} - \frac{\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)}$$

$$\kappa = -\frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}\alpha\left(x,y,t\right) + \frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}b\left(x,y,t\right)$$

$$\mu = -\frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}b\left(x,y,t\right) - \frac{1}{4}\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}b\left(x,y,t\right) - \frac{\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)} - \frac{\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)}$$

$$\lambda = -\frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}b\left(x,y,t\right) - \frac{1}{4}\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}b\left(x,y,t\right) + \frac{\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)} + \frac{\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)}$$

$$\nu = \frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}\alpha\left(x,y,t\right) - \frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}b\left(x,y,t\right)$$

$$\alpha = -\frac{\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)}$$

$$\beta = \frac{\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}N\left(z,x,y,t\right)}{4\,N\left(z,x,y,t\right)}$$

$$\tau = -\frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}\alpha\left(x,y,t\right) - \frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}b\left(x,y,t\right)$$

$$\pi = \frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}\alpha\left(x,y,t\right) + \frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial y}b\left(x,y,t\right)$$

$$\epsilon = \frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}\alpha\left(x,y,t\right) - \frac{1}{4}\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}b\left(x,y,t\right)$$

$$\gamma = \frac{1}{4}\sqrt{2}e^{(-b(x,y,t))}\frac{\partial}{\partial x}\alpha\left(x,y,t\right) + \frac{1}{4}\sqrt{2}e^{(-\alpha(x,y,t))}\frac{\partial}{\partial t}b\left(x,y,t\right)$$

Ricci Tensor Components

$$
\begin{aligned}
\Phi_{00} = {} & \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right)^2 + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right) - \frac{1}{2}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) \\
& + \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{2\, N\left(z,x,y,t\right)} \\
& + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} \alpha\left(x,y,t\right) + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} b\left(x,y,t\right) + \frac{1}{2}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial x \partial t} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} b\left(x,y,t\right) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{2\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial x \partial t} N\left(z,x,y,t\right)}{2\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{01} = {} & -\frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right) - \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) \\
& - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y} \alpha\left(x,y,t\right) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& + \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t} b\left(x,y,t\right) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{02} = {} & -\frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right)^2 - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) \\
& + \frac{1}{2}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} b\left(x,y,t\right)^2 + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} \alpha\left(x,y,t\right) \\
& + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} b\left(x,y,t\right) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} b\left(x,y,t\right) - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2} b\left(x,y,t\right) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}
\end{aligned}
$$

$$\Phi_{10} = -\frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\,(x,y,t)\, \frac{\partial}{\partial y}\alpha\,(x,y,t) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}\alpha\,(x,y,t)\, \frac{\partial}{\partial x}b\,(x,y,t) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\,(x,y,t)\, \frac{\partial}{\partial y}b\,(x,y,t) - \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y}\alpha\,(x,y,t)\, \frac{\partial}{\partial t}b\,(x,y,t)$$
$$- \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y}\alpha\,(x,y,t) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t}N\,(z,x,y,t)\, \frac{\partial}{\partial y}\alpha\,(x,y,t)}{4\,N\,(z,x,y,t)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}N\,(z,x,y,t)\, \frac{\partial}{\partial x}b\,(x,y,t)}{4\,N\,(z,x,y,t)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}N\,(z,x,y,t)\, \frac{\partial}{\partial y}b\,(x,y,t)}{4\,N\,(z,x,y,t)}$$
$$+ \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t}b\,(x,y,t) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y}N\,(z,x,y,t)\, \frac{\partial}{\partial t}b\,(x,y,t)}{4\,N\,(z,x,y,t)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y}N\,(z,x,y,t)}{4\,N\,(z,x,y,t)} + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t}N\,(z,x,y,t)}{4\,N\,(z,x,y,t)}$$

$$\Phi_{11} = \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\,(x,y,t)^2 - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\,(x,y,t)\, \frac{\partial}{\partial x}b\,(x,y,t) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}\alpha\,(x,y,t)\, \frac{\partial}{\partial y}b\,(x,y,t)$$
$$+ \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}\alpha\,(x,y,t)\, \frac{\partial}{\partial t}b\,(x,y,t) - \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}b\,(x,y,t)^2 + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2}\alpha\,(x,y,t) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}N\,(z,x,y,t)\, \frac{\partial}{\partial x}b\,(x,y,t)}{4\,N\,(z,x,y,t)}$$
$$+ \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}N\,(z,x,y,t)\, \frac{\partial}{\partial y}b\,(x,y,t)}{4\,N\,(z,x,y,t)} + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}N\,(z,x,y,t)\, \frac{\partial}{\partial t}b\,(x,y,t)}{4\,N\,(z,x,y,t)} - \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2}b\,(x,y,t) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2}N\,(z,x,y,t)}{4\,N\,(z,x,y,t)}$$

$$\Phi_{12} = \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\,(x,y,t)\, \frac{\partial}{\partial y}\alpha\,(x,y,t) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}\alpha\,(x,y,t)\, \frac{\partial}{\partial x}b\,(x,y,t) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\,(x,y,t)\, \frac{\partial}{\partial y}b\,(x,y,t) - \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y}\alpha\,(x,y,t)\, \frac{\partial}{\partial t}b\,(x,y,t)$$
$$+ \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y}\alpha\,(x,y,t) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t}N\,(z,x,y,t)\, \frac{\partial}{\partial y}\alpha\,(x,y,t)}{4\,N\,(z,x,y,t)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}N\,(z,x,y,t)\, \frac{\partial}{\partial x}b\,(x,y,t)}{4\,N\,(z,x,y,t)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}N\,(z,x,y,t)\, \frac{\partial}{\partial y}b\,(x,y,t)}{4\,N\,(z,x,y,t)}$$
$$+ \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t}b\,(x,y,t) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y}N\,(z,x,y,t)\, \frac{\partial}{\partial t}b\,(x,y,t)}{4\,N\,(z,x,y,t)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y}N\,(z,x,y,t)}{4\,N\,(z,x,y,t)} + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t}N\,(z,x,y,t)}{4\,N\,(z,x,y,t)}$$

$$\Phi_{20} = -\frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}\alpha\,(x,y,t)^2 - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\,(x,y,t)\, \frac{\partial}{\partial x}b\,(x,y,t) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}\alpha\,(x,y,t)\, \frac{\partial}{\partial y}b\,(x,y,t) - \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}\alpha\,(x,y,t)\, \frac{\partial}{\partial t}b\,(x,y,t)$$
$$+ \frac{1}{2}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}b\,(x,y,t)^2 + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}N\,(z,x,y,t)\, \frac{\partial}{\partial x}\alpha\,(x,y,t)}{4\,N\,(z,x,y,t)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}N\,(z,x,y,t)\, \frac{\partial}{\partial y}\alpha\,(x,y,t)}{4\,N\,(z,x,y,t)} - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2}\alpha\,(x,y,t)$$
$$+ \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}N\,(z,x,y,t)\, \frac{\partial}{\partial t}\alpha\,(x,y,t)}{4\,N\,(z,x,y,t)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}N\,(z,x,y,t)\, \frac{\partial}{\partial x}b\,(x,y,t)}{4\,N\,(z,x,y,t)} - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2}b\,(x,y,t) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}N\,(z,x,y,t)\, \frac{\partial}{\partial y}b\,(x,y,t)}{4\,N\,(z,x,y,t)}$$
$$- \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2}b\,(x,y,t) - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}N\,(z,x,y,t)\, \frac{\partial}{\partial t}b\,(x,y,t)}{4\,N\,(z,x,y,t)} + \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2}b\,(x,y,t) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2}N\,(z,x,y,t)}{4\,N\,(z,x,y,t)} - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2}N\,(z,x,y,t)}{4\,N\,(z,x,y,t)}$$

$$
\begin{aligned}
\Phi_{21} = {} & \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right) - \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) \\
& + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y} \alpha\left(x,y,t\right) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& + \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t} b\left(x,y,t\right) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right)^2 + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right) + \frac{1}{2}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) \\
& + \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{2\, N\left(z,x,y,t\right)} \\
& + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial y\right)^2} \alpha\left(x,y,t\right) + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial x\right)^2} b\left(x,y,t\right) - \frac{1}{2}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial x \partial t} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial y\right)^2} b\left(x,y,t\right) + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{2\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{\left(\partial t\right)^2} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial x\right)^2} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial x \partial t} N\left(z,x,y,t\right)}{2\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{\left(\partial t\right)^2} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & -\frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right)^2 - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right)^2 - \frac{1}{6}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) + \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} b\left(x,y,t\right)^2 \\
& - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{12\, N\left(z,x,y,t\right)} - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial x\right)^2} \alpha\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{12\, N\left(z,x,y,t\right)} - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial y\right)^2} \alpha\left(x,y,t\right) \\
& - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} \alpha\left(x,y,t\right)}{12\, N\left(z,x,y,t\right)} - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial x\right)^2} b\left(x,y,t\right) - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial y\right)^2} b\left(x,y,t\right) + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{6\, N\left(z,x,y,t\right)} \\
& + \frac{1}{6}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{\left(\partial t\right)^2} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial x\right)^2} N\left(z,x,y,t\right)}{12\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\left(\partial y\right)^2} N\left(z,x,y,t\right)}{12\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{\left(\partial t\right)^2} N\left(z,x,y,t\right)}{12\, N\left(z,x,y,t\right)}
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = {} & \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right)^2 + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right) - \frac{1}{2}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) \\
& + \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{2\, N\left(z,x,y,t\right)} \\
& - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} \alpha\left(x,y,t\right) - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} b\left(x,y,t\right) + \frac{1}{2}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial x \partial t} b\left(x,y,t\right) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} b\left(x,y,t\right) + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{2\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2} b\left(x,y,t\right) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial x \partial t} N\left(z,x,y,t\right)}{2\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_1 = {} & -\frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right) - \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) \\
& - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y} \alpha\left(x,y,t\right) + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& + \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t} b\left(x,y,t\right) + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t} N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_2 = {} & \frac{1}{6}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right)^2 - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right)^2 - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} \alpha\left(x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right) + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} \alpha\left(x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right) \\
& + \frac{1}{12}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} \alpha\left(x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} \alpha\left(x,y,t\right)}{12\, N\left(z,x,y,t\right)} + \frac{1}{6}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} \alpha\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} \alpha\left(x,y,t\right)}{12\, N\left(z,x,y,t\right)} \\
& - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} \alpha\left(x,y,t\right) - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} \alpha\left(x,y,t\right)}{12\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x} N\left(z,x,y,t\right) \frac{\partial}{\partial x} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} \\
& - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y} N\left(z,x,y,t\right) \frac{\partial}{\partial y} b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{1}{12}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} b\left(x,y,t\right) - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t} N\left(z,x,y,t\right) \frac{\partial}{\partial t} b\left(x,y,t\right)}{12\, N\left(z,x,y,t\right)} \\
& - \frac{1}{12}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2} b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2} N\left(z,x,y,t\right)}{12\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2} N\left(z,x,y,t\right)}{6\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2} N\left(z,x,y,t\right)}{12\, N\left(z,x,y,t\right)}
\end{aligned}
$$

$$\Psi_3 = \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\left(x,y,t\right) \frac{\partial}{\partial y}\alpha\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}\alpha\left(x,y,t\right) \frac{\partial}{\partial x}b\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\left(x,y,t\right) \frac{\partial}{\partial y}b\left(x,y,t\right) - \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y}\alpha\left(x,y,t\right) \frac{\partial}{\partial t}b\left(x,y,t\right)$$
$$+ \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y}\alpha\left(x,y,t\right) + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t}N\left(z,x,y,t\right) \frac{\partial}{\partial y}\alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}N\left(z,x,y,t\right) \frac{\partial}{\partial x}b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}N\left(z,x,y,t\right) \frac{\partial}{\partial y}b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)}$$
$$+ \frac{1}{4}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t}b\left(x,y,t\right) + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial y}N\left(z,x,y,t\right) \frac{\partial}{\partial t}b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{\partial x \partial y}N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial y \partial t}N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}$$

$$\Psi_4 = \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}\alpha\left(x,y,t\right)^2 + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}\alpha\left(x,y,t\right) \frac{\partial}{\partial x}b\left(x,y,t\right) - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}\alpha\left(x,y,t\right) \frac{\partial}{\partial y}b\left(x,y,t\right) + \frac{1}{2}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial x}\alpha\left(x,y,t\right) \frac{\partial}{\partial t}b\left(x,y,t\right)$$
$$+ \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}\alpha\left(x,y,t\right) \frac{\partial}{\partial t}b\left(x,y,t\right) - \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}N\left(z,x,y,t\right) \frac{\partial}{\partial x}\alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial t}N\left(z,x,y,t\right) \frac{\partial}{\partial x}\alpha\left(x,y,t\right)}{2\, N\left(z,x,y,t\right)}$$
$$- \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}N\left(z,x,y,t\right) \frac{\partial}{\partial y}\alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2}\alpha\left(x,y,t\right) - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}N\left(z,x,y,t\right) \frac{\partial}{\partial t}\alpha\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)}$$
$$- \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial x}N\left(z,x,y,t\right) \frac{\partial}{\partial x}b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)} - \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2}b\left(x,y,t\right) - \frac{1}{2}\, e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial x \partial t}b\left(x,y,t\right) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial}{\partial y}N\left(z,x,y,t\right) \frac{\partial}{\partial y}b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)}$$
$$- \frac{1}{4}\, e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial y)^2}b\left(x,y,t\right) - \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial}{\partial x}N\left(z,x,y,t\right) \frac{\partial}{\partial t}b\left(x,y,t\right)}{2\, N\left(z,x,y,t\right)} - \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial}{\partial t}N\left(z,x,y,t\right) \frac{\partial}{\partial t}b\left(x,y,t\right)}{4\, N\left(z,x,y,t\right)}$$
$$- \frac{1}{4}\, e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2}b\left(x,y,t\right) + \frac{e^{(-2\, b(x,y,t))} \frac{\partial^2}{(\partial x)^2}N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)} + \frac{e^{(-\alpha(x,y,t)-b(x,y,t))} \frac{\partial^2}{\partial x \partial t}N\left(z,x,y,t\right)}{2\, N\left(z,x,y,t\right)} + \frac{e^{(-2\, \alpha(x,y,t))} \frac{\partial^2}{(\partial t)^2}N\left(z,x,y,t\right)}{4\, N\left(z,x,y,t\right)}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"